

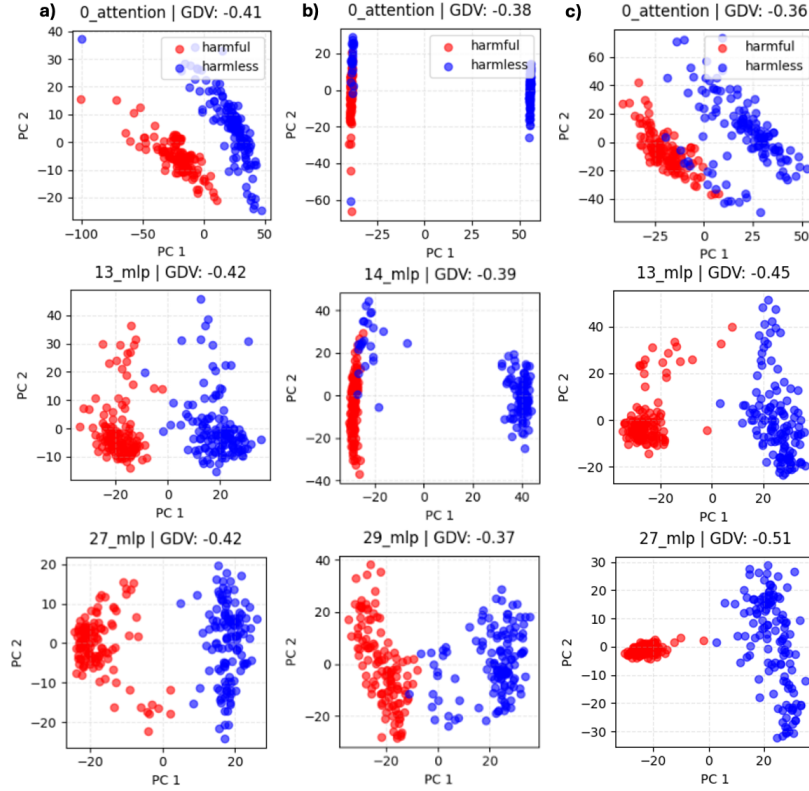


Fig. 1. Dimensionality-reduced residual activations using PCA, comparing harmful and harmless instructions across three different models and layers. **a)** Qwen2-1.5B-Instruct model (first, middle, last layers). **b)** Bloom-3b model (first, middle, last layers). **c)** Llama-3.2-3B-Instruct model (first, middle, last layers).

identify nonlinear relationships, offering deeper insights into the activation space.

UMAP and t-SNE consistently yielded clearer separations than PCA, as indicated by lower GDV scores across all models. For instance, the Qwen2-1.5B-Instruct model achieved nearly perfect cluster separability in its 18th layer, with distinct, compact clusters for harmful and harmless instructions (Fig. 2). These clusters, characterized by large inter-cluster and compact intra-cluster distances, provide a strong basis for probing classifier training.

Further analysis revealed evolving cluster morphologies across layers. Linear separability changed gradually, while UMAP and t-SNE highlighted dynamic variations in cluster shapes and the emergence of sub-clusters. These sub-clusters, as seen in Fig. 3, suggest additional nuanced features, potentially indicating the division of harmful instructions into finer sub-features.

These findings underscore that refusal behavior is a multidimensional and nonlinear phenomenon, necessitating advanced techniques for comprehensive analysis and interpretability.

Distinct refusal mechanisms across different model families

Unlike the localized functional architecture of the human brain, where specialized areas handle specific tasks, LLMs exhibit diverse strategies for embedding the distinction between harmful and harmless instructions. Refusal behavior

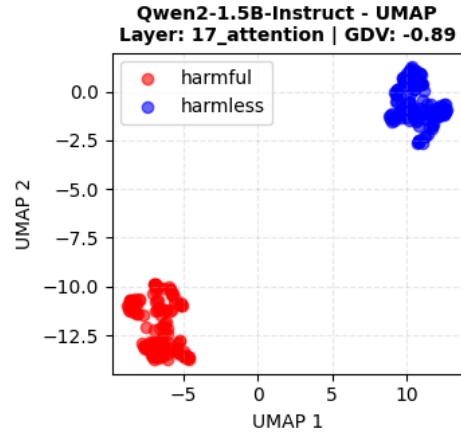


Fig. 2. Dimensionality-reduced residual activations of the Qwen2-1.5B-Instruct model at layer 18, visualized using UMAP, showing distinct clusters for harmful and harmless instructions.

varies across model families and layers, as visualized in Fig. 4.

Qwen2 Models. In the Qwen2 family, refusal behavior is primarily encoded in early layers. PCA reveals that the refusal feature emerges within the first few layers, with the 0.5B model peaking at the first layer and the 1.5B model at the

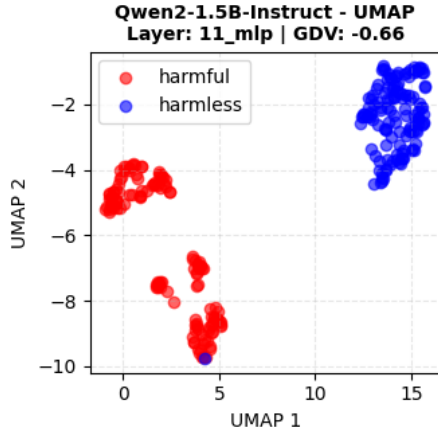


Fig. 3. Dimensionality-reduced residual activations of the Qwen2-1.5B-Instruct model at layer 11, showing the emergence of sub-clusters for harmful instructions.

fourth. Early layers integrate multiple principal components to define the refusal direction, while later layers refine it along the first principal component. This progression leads to stable inter-class distances and declining intra-class distances, resulting in compact harmful clusters. UMAP and t-SNE further confirm robust separability after the fifth layer.

Bloom Models. The Bloom architecture exhibits a distinct refusal mechanism, with a tendency to misclassify harmless instructions as harmful. The refusal direction aligns with the first principal component or main UMAP/t-SNE direction from the initial layer, where instructions are classified in a near-binary manner. GDV peaks in the fifth and sixth layers, but subsequent layers show weakened discrimination as intra-class distances increase and inter-class distances decrease. UMAP visualizations reveal sub-clusters within harmless instructions.

Llama Models. Llama models display a gradual refinement of refusal behavior. In early layers, harmful and harmless embeddings are mixed, indicating weak differentiation. Refusal strength intensifies in deeper layers, with Llama-3.2-3B-Instruct achieving compact clusters for harmful instructions and widespread embeddings for harmless ones, demonstrating stronger recognition. Conversely, the smaller Llama-3.2-1B-Instruct, with fewer hidden dimensions, shows more mixed embeddings, consistent with the superposition hypothesis. Nonlinear methods, such as t-SNE and UMAP, corroborate the emergence of clearer refusal representations in middle layers.

These findings reveal that refusal behavior is universally present but uniquely mediated across different architectures, reflecting varied strategies for harmful content differentiation.

DISCUSSION

Refusal behavior, enabling differentiation between harmful and harmless instructions, was consistently observed across all six models tested, corroborating previous findings [13].

Our analysis highlights that refusal mechanisms vary significantly across model families, reflecting architecture-specific

strategies for harmful content detection. Qwen2 models primarily encode refusal features in early layers, achieving stable inter-class distances and compact intra-class structures. Bloom models exhibit peak separability in intermediate layers but demonstrate weaker discrimination in later layers, often misclassifying harmless instructions. Conversely, Llama models refine refusal gradually across layers, with stronger separability emerging in deeper layers.

Contrary to the assumption that refusal resides in a simple one-dimensional linear subspace [13], our findings reveal a more complex, nonlinear structure. Using PCA, UMAP, and t-SNE, we visualized latent space activations and found that nonlinear methods consistently outperformed PCA in separating harmful and harmless embeddings, as evidenced by the separability metric GDV.

These results align with recent studies on jailbreak attacks, which indicate that nonlinear features, rather than universal linear ones, drive the success of such attacks [29]. Although focused on jailbreak mechanisms, these findings suggest a potential connection between jailbreak features and the refusal and harmfulness features, given the critical role of refusal in determining whether a model responds or declines.

Furthermore, the phenomenon of "alignment faking," where LLMs pretend to align with training objectives while resisting preference modifications [30], raises concerns about the robustness of model alignment processes. Persistent alignment faking could lock in model preferences, posing challenges for future fine-tuning efforts.

These insights underscore the importance of nonlinear interpretability techniques for understanding refusal behavior and its implications for alignment, robustness, and ethical AI deployment.

CONCLUSION

This study investigates refusal behavior in large language models (LLMs) and provides new insights into its mechanisms and characteristics. By analyzing six LLMs across three architectural families, we demonstrate that refusal behavior is a universal phenomenon but exhibits architecture-specific patterns. Contrary to prior assumptions that refusal mechanisms are linear and confined to single activation directions, our findings reveal that they are inherently nonlinear and multidimensional. Using advanced dimensionality reduction techniques such as UMAP and t-SNE, we uncover richer and more complex activation patterns than those detected through traditional linear methods like PCA.

The results emphasize the importance of nonlinear interpretability methods in understanding and improving model alignment. Refusal mechanisms vary significantly across architectures, with Qwen models encoding refusal early, Bloom models demonstrating intermediate-layer strengths, and Llama models refining refusal behavior in deeper layers. These differences highlight the diverse strategies employed by LLMs to distinguish harmful and harmless prompts, which has implications for designing safer, more robust AI systems.

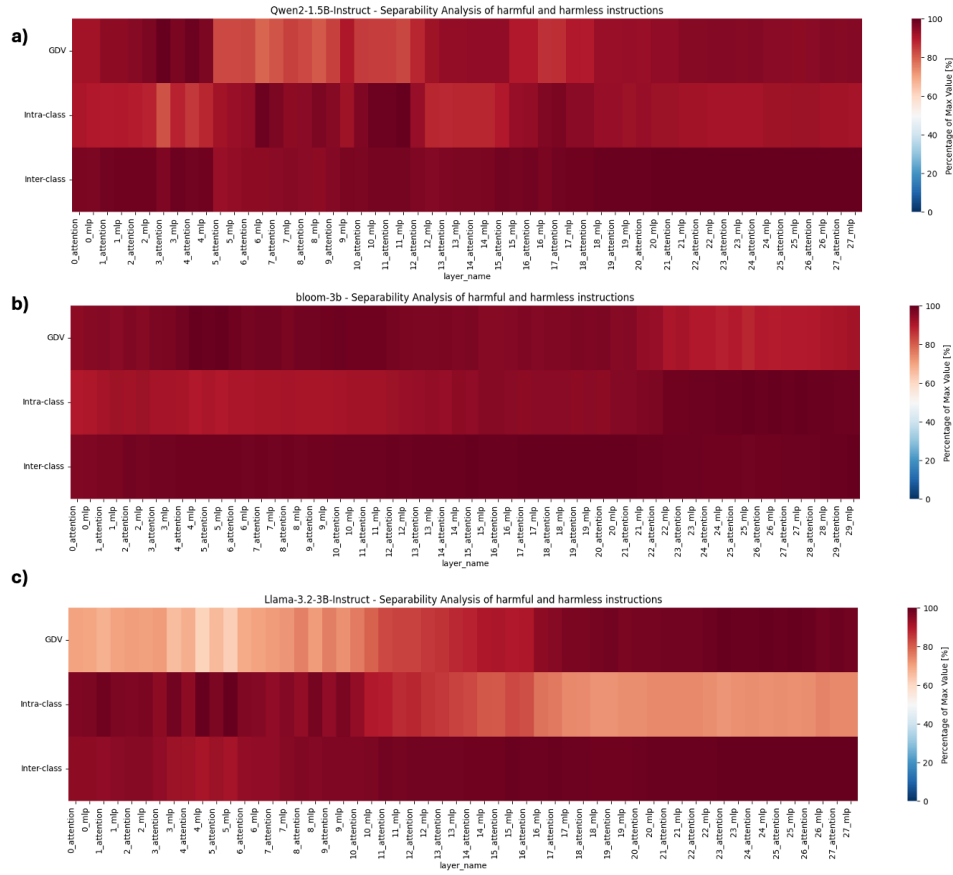


Fig. 4. GDV, intra-class distance (compactness of harmful and harmless clusters), and inter-class distance (separation between clusters) of the dimensionality-reduced embeddings using PCA. **a)** Qwen2-1.5B-Instruct demonstrates early layer dominance of the refusal feature. **b)** Bloom-3b shows peak separability at early to intermediate layers but weaker discrimination in later layers. **c)** Llama-3.2-3B-Instruct displays progressively stronger separation.

Future research should explore the transferability of non-linear refusal probes across model families and examine their interplay with jailbreak attacks and other adversarial strategies. Understanding refusal behavior’s nuanced and nonlinear nature will be pivotal in enhancing the alignment, transparency, and ethical deployment of LLMs.

AUTHOR CONTRIBUTIONS

FH, PK and AS conceptualized the study and designed the study. FH conducted the experiments. PK, AM and AS provided guidance, supervision, and feedback on the research design and interpretation of results. All authors wrote, reviewed and approved the final manuscript.

CODE AVAILABILITY

All results and the code are published on a public GitHub repository [31] under the MIT license.

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